University of Rwanda
CST
Mobile Application Development.
Computer Engineering
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#LAB 4

Consuming APIs in Flutter LAB 4.

Post manager is a flutter mobile application that allow staff of small media company to view, create, edit and delete posts through a REST API, the backend is simulated using JSONPlaceholder, a free fake API for testing and prototyping.

Features	Description.
View all posts	Fetches and lists 100 posts from the API.
Search posts	Real-time search by title or content with keyword highlighting
Filter by user	chips to filter posts by user ID
view post detail	Collapsing header with full post content.
Create a post	Form with validation, sends POST request to API
Edit a post	pre-filled form, unsaved-changes guard, sends PUT request.
Delete a post	swipe-to-delete or button, with confirmation dialog
pull to refresh	Drop down to reload posts from API.
Error handling	~~Navigation~~ network placeholder cards while data is loading

New dependencies

http : Sends HTTP requests (GET, POST, PUT, DELETE), lightweight official Dart package.

Error Handling

SocketException : NO Internet connection
TimeOutException : Server too slow.

which part of UI depends on the FUTURE?
The entire body of posts list screen ~~is~~ driven by Future Builder. Nothing in the list is rendered until the future resolves once resolved, the posts are stored in _allposts and _filtered.

~~Set~~
Setup for Android to connect on Internet:
android/app/src/main/AndroidManifest.xml.
< Uses permission android:name="android.permission.INTERNET"/>

Question I discussed:

1. Which dependencies did you use and why?
2. How can you handle API exceptions, network errors, invalid data?
3. Explain the Future Builder widget and which part of the UI depends on it.